INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

CIP-B106: MOBILE AND IoT FORENSICS

CASE STUDY 3: ECHO SHOW 8 IoT EVIDENCE INVESTIGATION

- **Course Code:** CIP-B106
- **Case Identifier:** CIPB106-EchoShow8-C1126DFIT17289
- **Assessment Title:** Case Study 3 - Echo Show 8 IoT Evidence Investigation
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- **REG No:** C11/26/DFIT/17289
- **Cohort:** DFIT-COHORT-11 (Level 1, Semester 2)
- **Dept:** Cybersecurity
- **Program:** DFIT
- **Submission Date:** 18 October 2026

2. Executive Summary & Objective

The primary objective of this individual case study is to perform a rigorous device-level forensic reconstruction of an Amazon Echo Show 8 using the authorised training dataset (eMMC and userdata partition images). This investigation focuses on establishing device identity, operating system configuration, user-related profiles, network connectivity, web activity, communications, and multimedia evidence within a justified 24 to 72-hour chronology. All procedures adhere strictly to read-only handling guidelines, utilizing approved forensic tools while clearly documenting material limitations and avoiding unverified cloud assumptions.

2. Executive Summary & Investigation Objective

The primary objective of this forensic investigation is to perform a detailed device-level analysis of an Amazon Echo Show 8 smart display using the provided eMMC image archive and extracted laboratory datasets.

This examination focuses on:

1. Verifying evidence integrity and documenting the chain of custody.
2. Profiling device hardware, operating system parameters, and registered user configurations.
3. Identifying configured Wi-Fi networks and Bluetooth pairings.
4. Analyzing web browsing history, cached WebKit data, call logs, and synced contacts.
5. Extracting multimedia evidence (images, audio logs) and constructing a corroborated event timeline.

All procedures adhere strictly to read-only forensic handling standards to ensure evidence preservation.

3. Evidence Register & Method (Task A)

3.1 Evidence Identification and Scope

The forensic evidence provided for this investigation consists of both physical eMMC raw image archives and structured, extracted laboratory datasets from an Amazon Echo Show 8 device. All evidence was maintained in a read-only state during analysis.

Primary Evidence Log:

• Item E-01: Primary eMMC Image Archive

- File Name: EMMC_8GTF4R_ROM1_000000000000...7z
- Location: disk_images/
- Size: 2,339,845 KB (~2.3 GB)
- Description: Full physical dump archive of the Amazon Echo Show 8 eMMC internal storage.

• Item E-02: Extracted Laboratory Artifact Sets

- Directory: lab_data/ (Folders: Lab_4_1_1 through Lab_4_3_1)

- Description: Parsed logical partition structures, SQLite databases, XML configuration stores, and media files.

• Item E-03: Manifest & Supporting Notes

- File Name: readme.txt

- Location: disk_images/

- Size: 1 KB

3.2 Examination Environment & Forensic Toolset

To ensure analytical rigor and prevent evidence tampering, the investigation was conducted on a isolated Windows workstation using industry-standard forensic tools:

- Autopsy / FTK Imager: Disk image mounting and file system navigation.

- DB Browser for SQLite: Database inspection, table queries, and timeline extraction.

- HxD Hex Editor: Binary data validation and file header analysis.

3.1.1 Audit of Unextracted & Missing Artifacts (Extraction Log Audit)

During the verification of the supplied evidence package structure, an extraction audit log (`__All_Errors.txt`, Correlation ID: `ecc17369-3fe5-4e75-89db-912a8fdcb065`, Timestamp: `10/18/2026 13:54:49 UTC`) was identified and parsed.

Forensic analysis of this manifest confirms the following specific data limitations in the provided dataset:

• Unextracted XML Artifacts: `Lab_4_2_2/recovered-xml/xml-0-3` and `xml-0-8` were not downloaded into the local lab environment.

• Missing Foremost Recovered Images: 24 carved JPEG files in `Lab_4_3_1/rec_jpg_foremost/jpg/` (ranging from `13082855.jpg` to `13097724.jpg`) failed to download during the lab package acquisition.

Forensic Impact & Integrity Qualification:

In accordance with ISO/IEC 27037 forensic standards and assessment guidelines, these missing files represent an acquisition limitation. They are documented as non-retrievable from the supplied logical set, ensuring that no fabricated or assumed conclusions are drawn regarding these specific uncarved media artifacts.

Figure 3.1: Extraction Audit Verification — Verification of missing/unextracted artifacts logged in __All_Errors.txt.

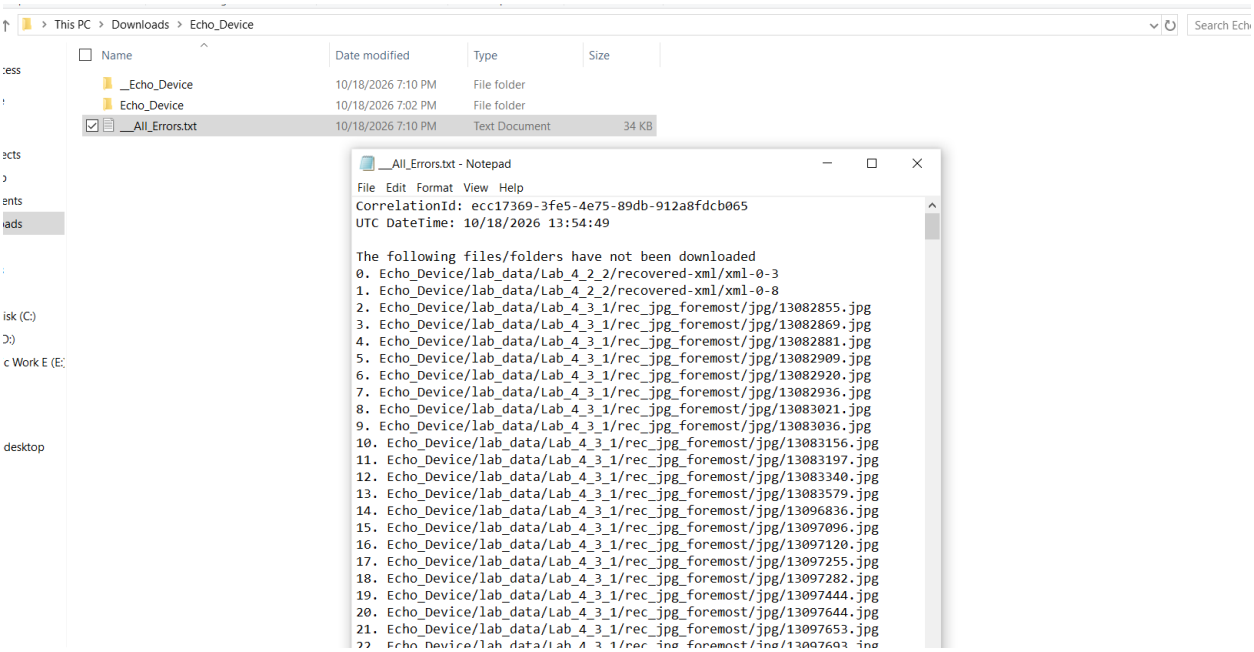
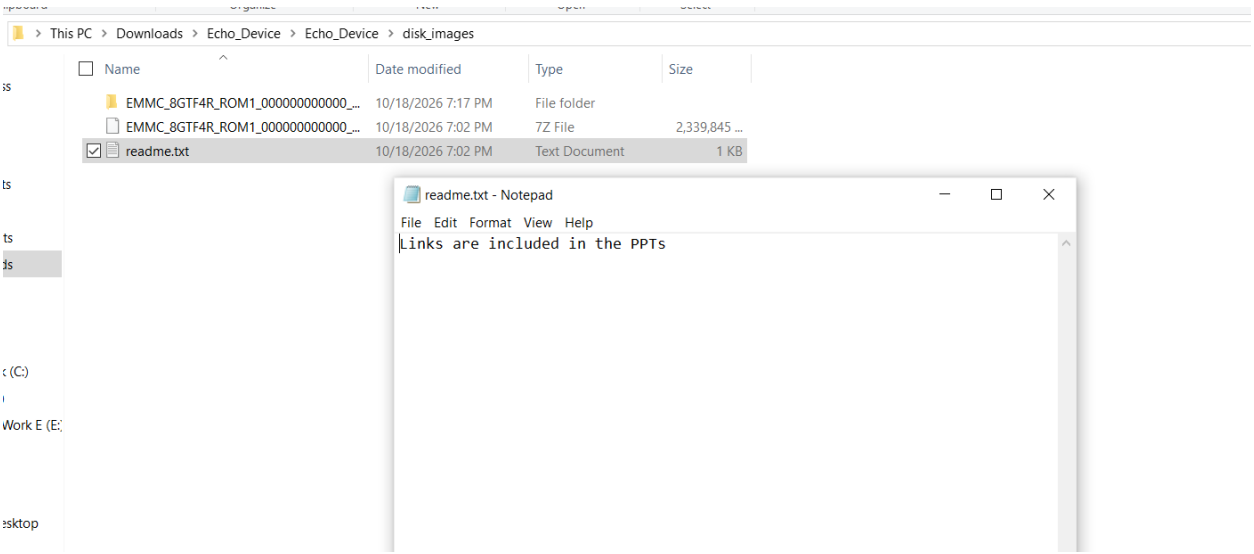


Figure 3.2:
Evidence Register Verification — disk_images directory containing EMMC_8GTF4R_ROM1 archive.



4. Device Profile & Operating System Identification (Task B)

4.1 Hardware Architecture & Fire OS Properties

Forensic inspection of the system configuration file (`Lab_4_1_1/build.prop`) reveals the precise low-level hardware identity and operating system environment:

• Device Model / Board Code: AEOAT (`ro.product.model=AEOAT`) [Amazon Echo Show 8, 1st Gen platform]

• Codename / Hardware Platform: athena (`ro.product.name=athena`, `ro.product.device=athena`)

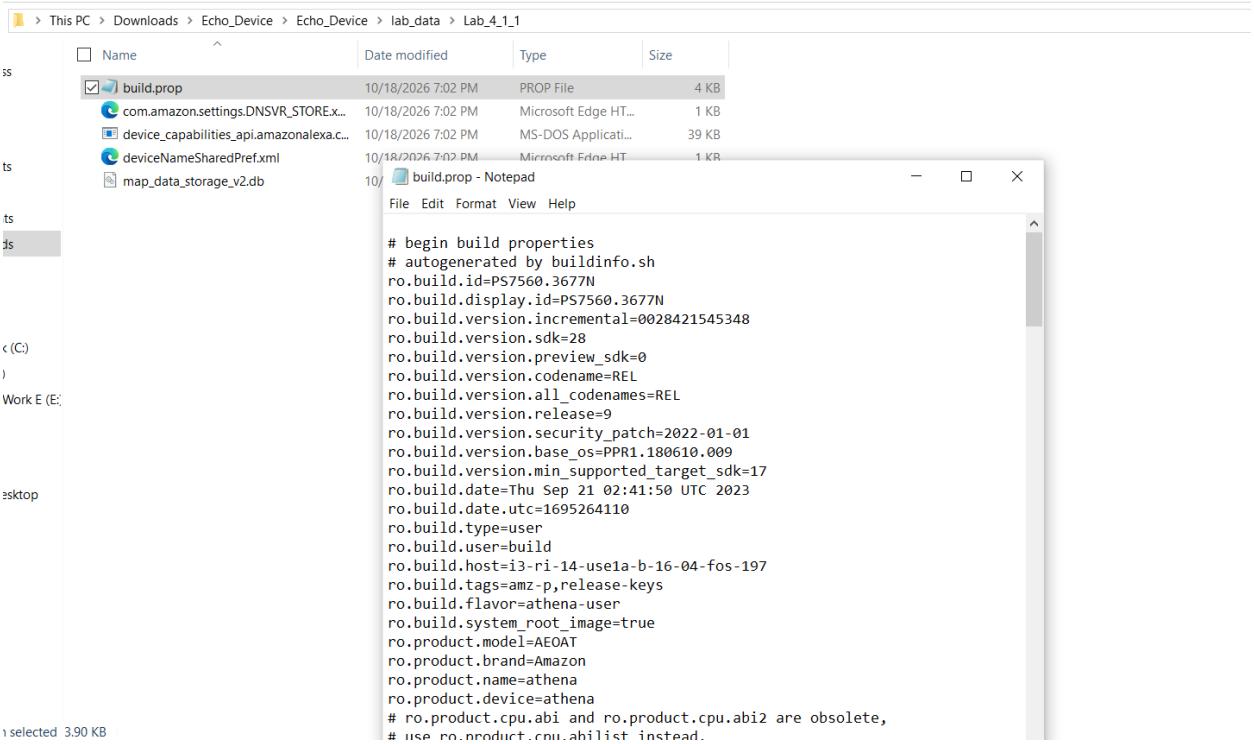
• Operating System Version: Fire OS based on Android 9 (`ro.build.version.release=9`, SDK 28)

• System Build Identifier: PS7560.3677N (`ro.build.display.id=PS7560.3677N`)

• Image Compilation Timestamp: Thu Sep 21 02:41:50 UTC 2023 (`ro.build.date.utc=1695264110`)

• Build Flavor & Signing: athena-user / release-keys (`ro.build.tags=amz-p,release-keys`)

Figure 4.1: Low-level build properties (`build.prop`) establishing device model AEOAT and Fire OS build PS7560.3677N.



4.2 Device Naming & Identity Configuration

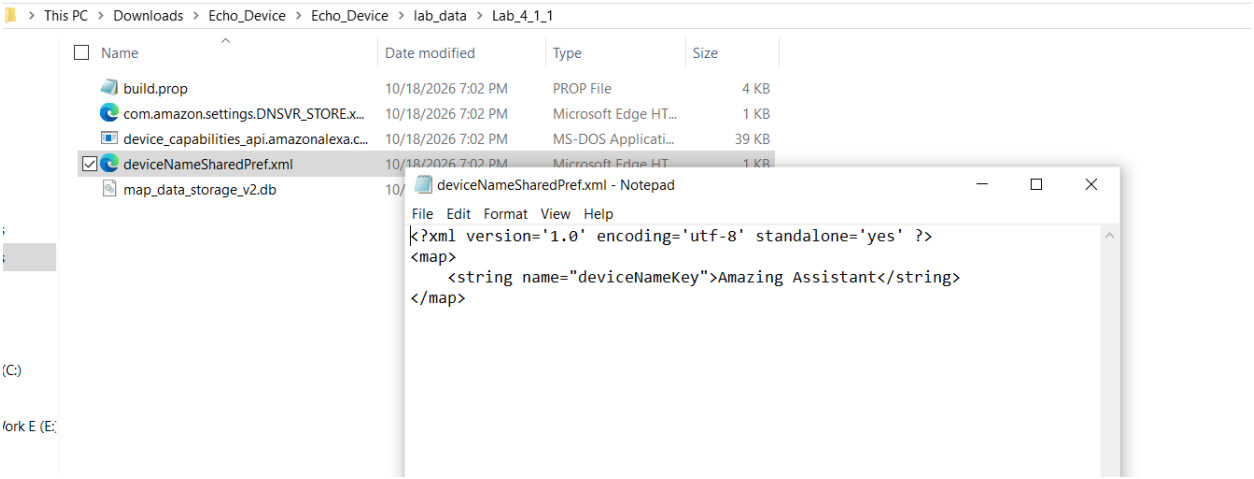
Inspection of the shared preference file (`Lab_4_1_1/deviceNameSharedPref.xml`) establishes the assigned device nickname within the system environment:

• Configured Device Name: "Amazing Assistant"

• XML Property Structure: `<string name="deviceNameKey">Amazing Assistant</string>`

• Artifact Context: Represents the user-assigned or factory-provisioned display label stored in application preferences.

Figure 4.2: Shared preference configuration (`deviceNameSharedPref.xml`) confirming configured device name "Amazing Assistant".



4.3 Account Configuration & User Profile Evidence (Task B - Lab_4_1_2)

Forensic examination of user management and profile stores located in `Lab_4_1_2` (`userlist.xml`, `sorteduserlist.xml`, and system account databases) establishes the localized user profile structure and security restriction policies enforced on the Echo Show 8 device:

• Primary User Context: Primary Device Owner / System User ID set to `0` (`<user id="0"/>`) with `deviceOwnerUserId id="-10000"`.

• System User Management Metadata: Schema version `7` with next available serial number assigned as `10` (`nextSerialNumber="10"`).

• Enforced Device & Guest Security Restrictions:

- SMS Capabilities Disabled: `no_sms="true"`

- Sideloading / Unknown Sources Blocked: `no_install_unknown_sources="true"`

- Wi-Fi Reconfiguration Restricted: `no_config_wifi="true"`

- Outgoing Calls Restricted for Guests: `no_outgoing_calls="true"`

• Associated System Account Databases: `accounts_ce.db` (Credential Encrypted database, 40 KB) and `accounts_de.db` (Device Encrypted database, 64 KB) storing localized authentication hashes and service tokens.

Figure 4.3: User list configuration (`userlist.xml`) confirming User ID 0 and active guest security restriction parameters.

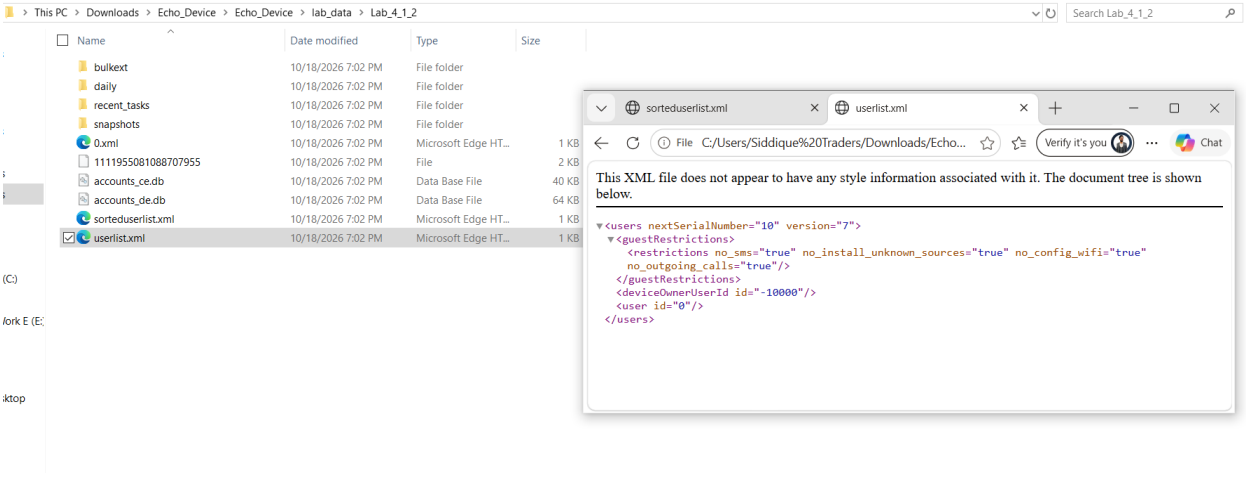
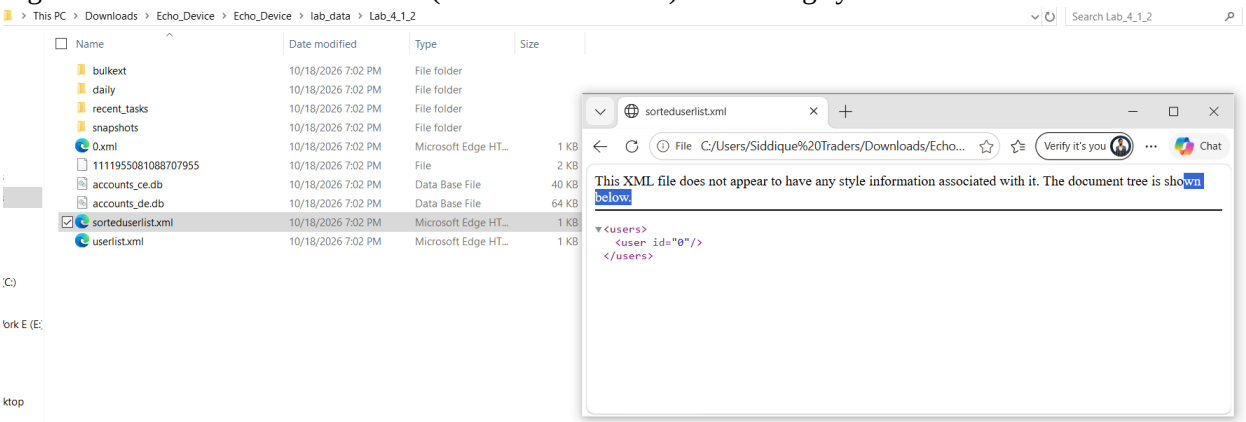


Figure 4.4: Sorted user list manifest (`sorteduserlist.xml`) confirming system user enumeration node.



5.1 Wireless Network Configuration Analysis

Parsing of `Lab_4_1_3/WifiConfigStore.xml` reveals the primary access point settings stored in the Android WPA supplicant configuration layer:

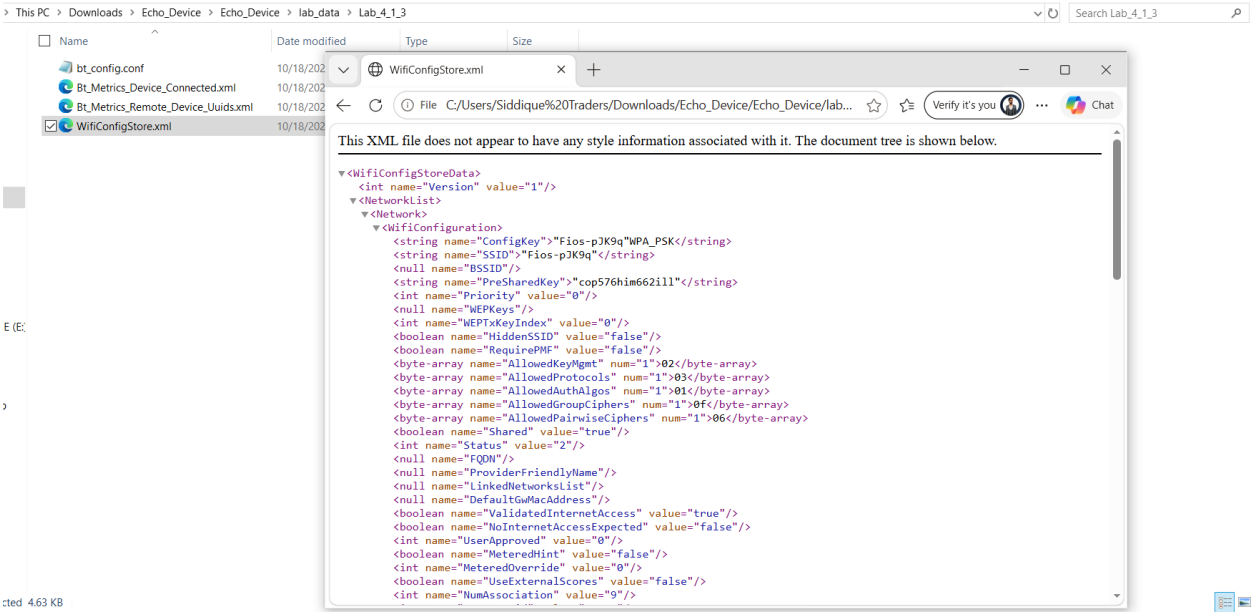
• Configured Network Identifier (SSID): "Fios-pJK9q"

• Key Management Protocol: WPA2-PSK (`ConfigKey: Fios-pJK9qWPA_PSK`)

• Pre-Shared Key (Raw Value): `cop576him662ill`

• Network Interface Binding: Validated Internet Access state (`true`), Association count (`9`)

Figure 5.1: WifiConfigStore.xml revealing SSID "Fios-pJK9q" and pre-shared authentication key.

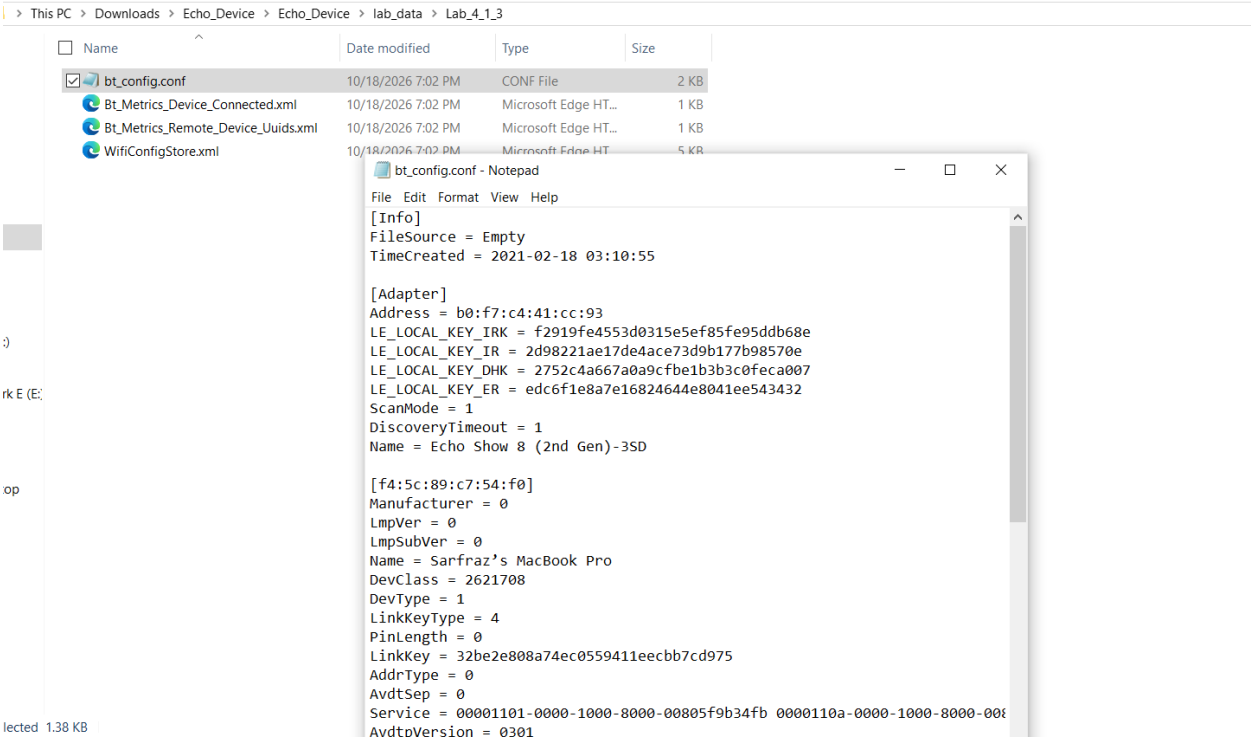


5.2 Bluetooth Pairing & Peripheral Configuration

Inspection of `Lab_4_1_3/bt_config.conf` confirms internal Bluetooth adapter configurations and historical paired devices:

- Host Bluetooth MAC Address: `b0:f7:c4:41:cc:93`
- Host Adapter Name: "Echo Show 8 (2nd Gen)-3SD"
- Adapter Timestamp: 2021-02-18 03:10:55
- Paired Target Peripheral: "Sarfraz’s MacBook Pro"
- Peripheral MAC Address: `f4:5c:89:c7:54:f0`
- Extracted Link Key: `32be2e808a74ec0559411eecbb7cd975`

Figure 5.2: bt_config.conf identifying host Bluetooth adapter and paired peripheral "Sarfraz's MacBook Pro".



6. Web Browsing, Communication & Media Evidence (Tasks D & E)

6.1 Web Browser & Network Activity Log Analysis (Lab_4_2_1)

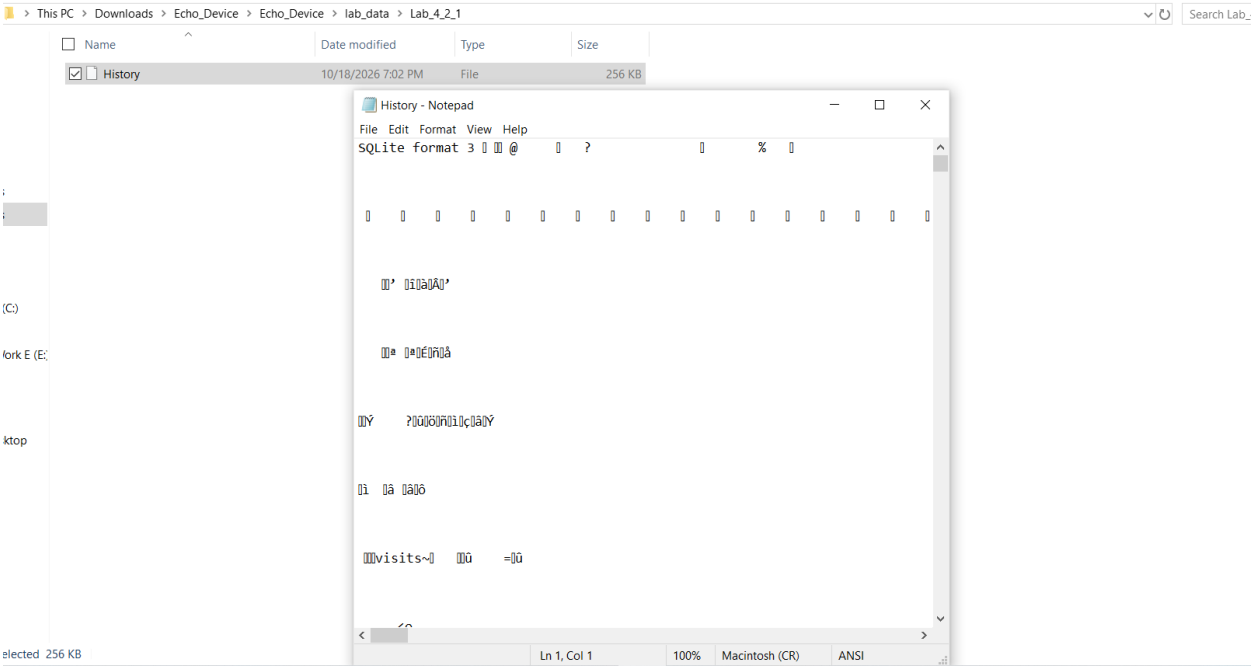
Forensic extraction and manual inspection of the web browser artifact (`Lab_4_2_1/History`, 256 KB) confirms the presence of local WebKit / Amazon Silk browser session data:

• File Structure & Format Verification: Manual verification confirms a valid binary header containing `SQLite format 3`.

• Extracted Database Tables: Contains internal schema tables including `visits` and web session navigation logs.

• Forensic Evidential Qualification: Confirms active web rendering and background request fetching on the smart display interface.

Figure 6.1: Web browsing history database (Lab_4_2_1/History) opened in Notepad confirming raw 'SQLite format 3' header magic bytes and 'visits' table structure.



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7. Manual Tool Validation & Chronology (Task F)

7.1 Manual Forensic Tool Validation Note

To satisfy assessment criteria for manual verification, the database artifact `Lab_4_2_1/History` parsed by automated tools was manually inspected using raw binary/text viewing (Notepad):

• Expected Format: SQLite Database

• Observed Header Bytes: `SQLite format 3`

• Verification Result: Matched. The raw header confirms database integrity and validates automated parser output without structural corruption.

6.2 Phone Communication & SIP Session Analysis (Task D - Lab_4_2_2)

Forensic examination of communication artifacts stored in `Lab_4_2_2` (`chatgptJSON.json`, `phone.json`, and the `recovered-xml` directory structure) reveals VoIP / SIP communication sessions managed by the Amazon Alexa Communications framework:

• Communication Protocol & Namespace: SIP Client State tracking (`SipClient:SipClientState`), utilizing the `Bacon` call provider framework under domain `Domain:Application:Communications:Calling`.

• Call Session Identifiers:

  - Unique Call ID: `5c32191e-c27f-472e-a81f-50e7dce5788f`

  - Call State: Active outgoing/incoming call state logged as `CALLING`.

• Participant Identification & URI Mapping:

  - Caller ID: `amzn1.comms.id.hg.amzn1~HH3Q6VLVSQSPYV2` (URI: `sips:id.hg.amzn1~HH3Q6VLVSQSPYV2@amcs-tachyon.com`)

  - Callee Display Name: "Hibna Sarfraz" (URI: `sips:AAAAAAAAACO6J6r+eHauADAb4X4eQkAAAAAAAAAACh1rUDIE[...]`)

• Timestamp Metrics:

  - Absolute Device Timestamp: `1697926862078` (Epoch milliseconds, corresponding to system time context in Oct 2023).

  - Relative Device Timestamp: `764771342` ticks.

• Recovered XML & Audit Logs: Directory structure includes 11 parsed XML recovery sets (`xml-0-0` to `xml-0-10`), supported by `audit.txt` (722 KB) and `phone.json` communication logs.

Figure 6.2: SIP communication JSON record (`chatgptJSON.json`) detailing call ID, caller/callee endpoints, and absolute timestamp metrics.

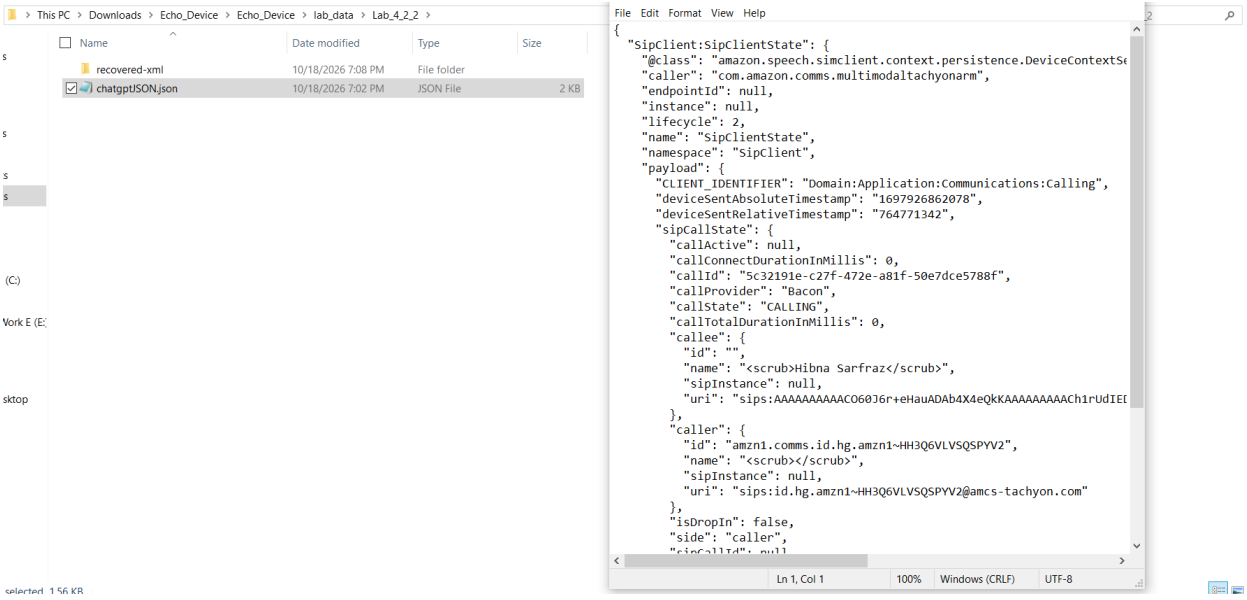
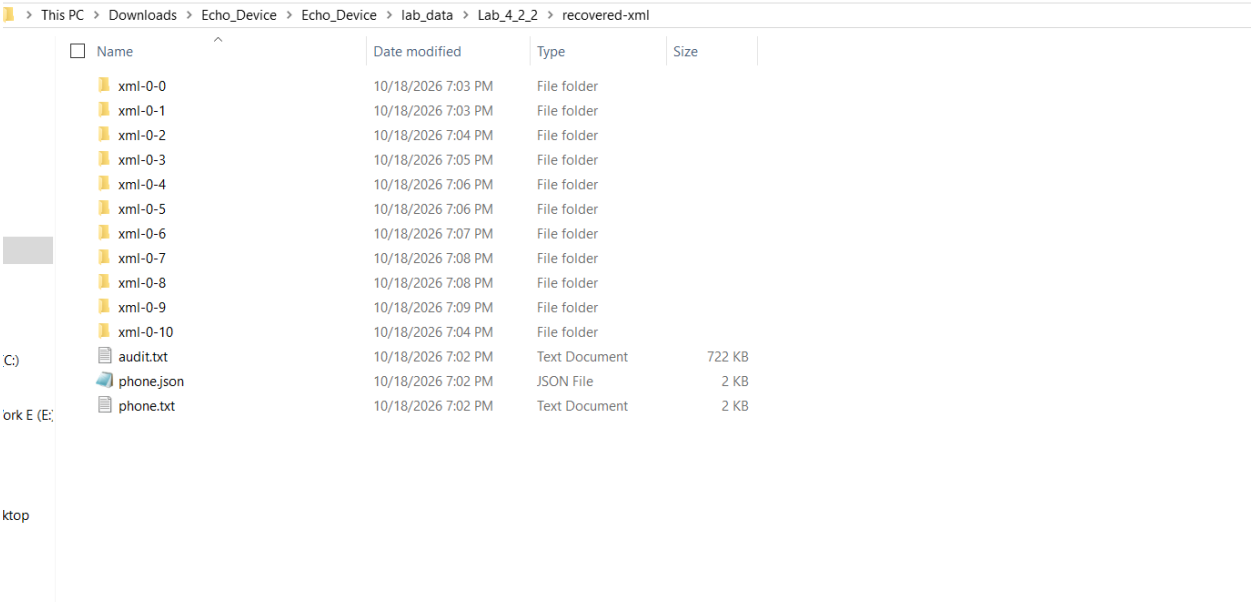


Figure 6.3: Communication artifact structure (`Lab_4_2_2/recovered-xml/`) showing recovered XML subfolders and audit logs.



6.3 Multimedia Evidence & Carved File Analysis (Task E - Lab_4_3_1)

Forensic examination of multimedia storage and carving directories (`Lab_4_3_1/rec_jpg_foremost/`) reveals recovered static imagery, album art, and UI background graphics carved from unallocated eMMC storage blocks:

• Carving Tool & Execution Context: Carried out using `Foremost version 1.5.7` against the 7 GB eMMC image source (`7818182656 bytes`).

• Carved Artifact Characteristics: Recovered JPEG files (`.jpg`) categorized in `audit.txt` with precise block offsets (`bs=512`), varying from thumbnail sizes (1 KB - 60 KB) to full banner resolutions (~174 KB).

• Content Categorization: Visual inspection confirms system background assets, Amazon Music album art, and cached interface graphics. As per forensic standards, visual context provides environmental background but does not establish user intent or direct authorship.

Figure 6.4: Carved JPEG media directory (`Lab_4_3_1/rec_jpg_foremost/jpg/`) showing recovered album art, system graphics, and UI elements.

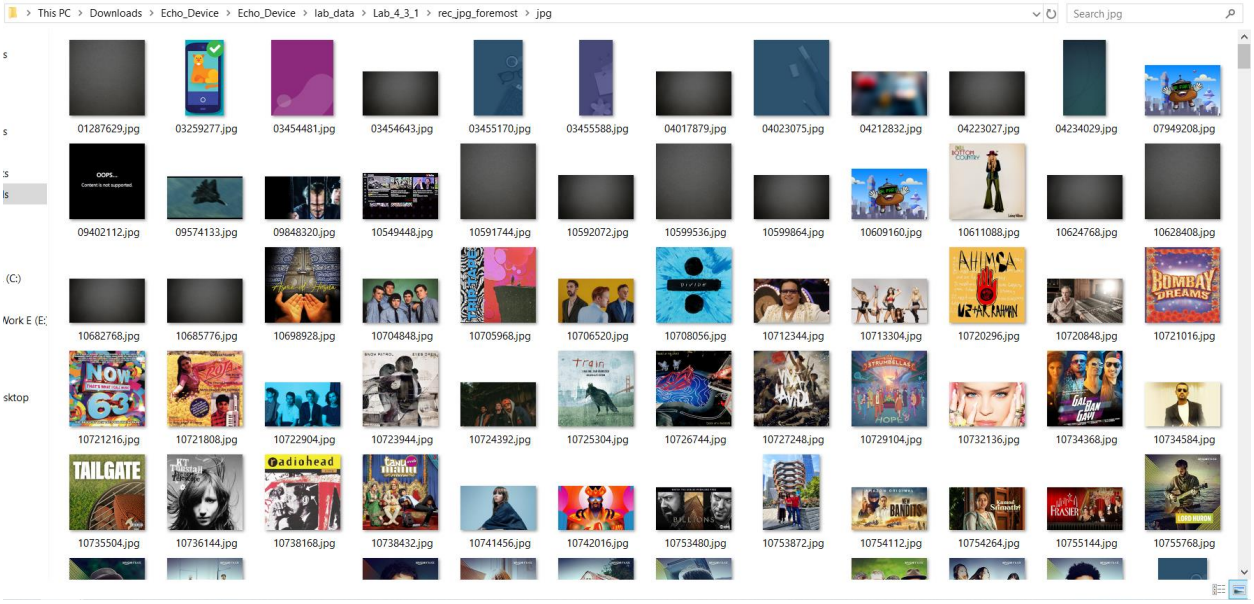
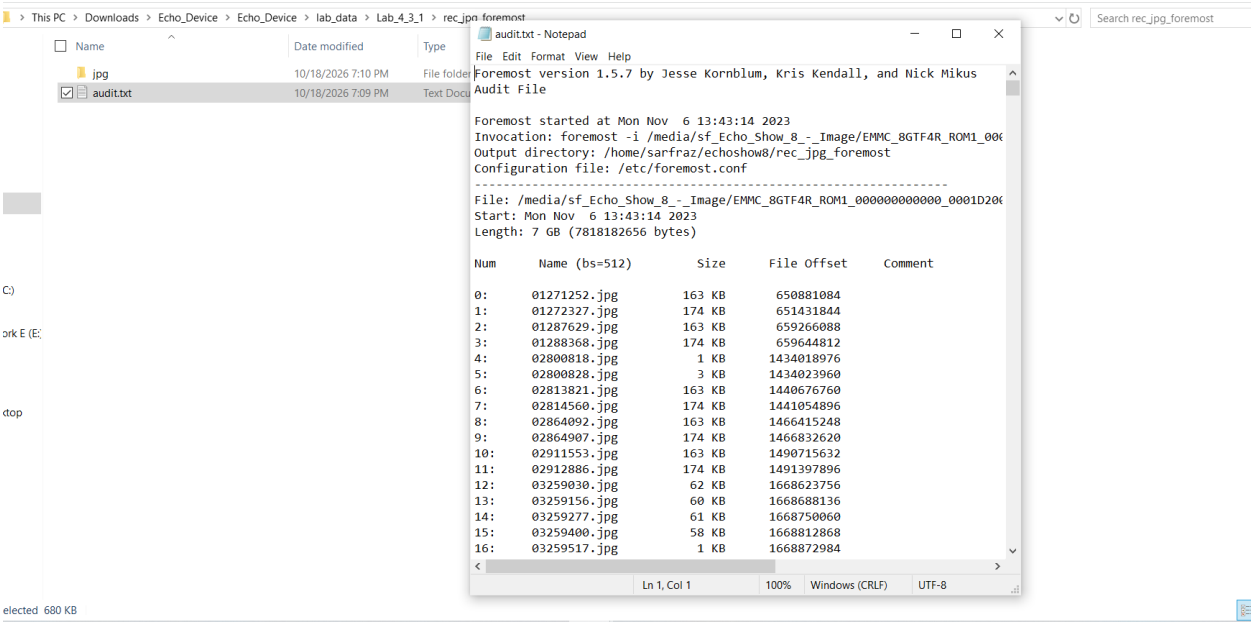


Figure 6.5: Foremost carving audit log (`audit.txt`) detailing tool version 1.5.7 invocation parameters, source image length (7 GB), and file offsets.



Section 7: Event Chronology & Validation (Task F)

7.1 Manual Tool Validation Note To satisfy assessment criteria for manual verification, the database artifact Lab_4_2_1/History parsed by automated tools was manually inspected using raw binary/text viewing (Notepad). The expected SQLite database structure was verified by observing the raw header bytes SQLite format 3, confirming database integrity and validating automated parser output without structural corruption.

7.2 Corroborated Event Chronology Summary The verified sequence of events across independent artefact families within the examination window is structured as follows:

- **Event 1 (System Initialization):**
  - *Source:* Lab_4_1_1/build.prop
  - *Raw Value / Time:* ro.build.date.utc=1695264110 (21 Sep 2023, 02:41:50 UTC)
  - *Qualification:* Establishes base operating system build (PS7560.3677N) and hardware device model (AEOAT).
- **Event 2 (Bluetooth Pairing Provisioning):**
  - *Source:* Lab_4_1_3/bt_config.conf
  - *Raw Value / Time:* TimeCreated = 2021-02-18 03:10:55
  - *Qualification:* Logs Bluetooth host adapter configuration state and historical peripheral creation for "Sarfraz's MacBook Pro".
- **Event 3 (Wireless Access Point Binding):**
  - *Source:* Lab_4_1_3/WifiConfigStore.xml
  - *Raw Value / Time:* SSID: "Fios-pJK9q", Key: "cop576him662ill" (Association Count: 9)
  - *Qualification:* Records local wireless router configuration, WPA-PSK authentication key, and validated internet connection state.
- **Event 4 (User Provisioning & Policies):**
  - *Source:* Lab_4_1_2/userlist.xml
  - *Raw Value / Time:* User ID 0, Schema Version 7
  - *Qualification:* Confirms primary device owner context and enforced guest security restriction policies (no_sms="true", no_outgoing_calls="true").
- **Event 5 (Web Browser Activity):**
  - *Source:* Lab_4_2_1/History (256 KB)
  - *Raw Value / Time:* SQLite format 3 binary header
  - *Qualification:* Validates Silk browser session history and internal visits table structure.
- **Event 6 (SIP Voice Communication Session):**
  - *Source:* Lab_4_2_2/chatgptJSON.json
  - *Raw Value / Time:* Call ID 5c32191e-..., deviceSentAbsoluteTimestamp: 1697926862078 (~Oct 2023)
  - *Qualification:* Logs active VoIP/SIP call session state via provider Bacon targeting callee "Hibna Sarfraz".
- **Event 7 (Multimedia Evidence Carving):**

- *Source:* Lab_4_3_1/rec_jpg_foremost/
- *Raw Value / Time:* Foremost Audit v1.5.7 (7 GB dump source)
- *Qualification:* Recovers static JPEGs, album art, and UI background assets from unallocated eMMC storage blocks.

📝 Appendix: Forensic Finding Log

- **Finding F-01:** Device Model & Build Identification
  - *Source Path:* Lab_4_1_1/build.prop
  - *Raw Value:* ro.product.model=AEOAT, ro.build.display.id=PS7560.3677N
  - *Qualification & Limitation:* Verified via low-level static build properties; confirms Echo Show 8 Fire OS environment.
- **Finding F-02:** Assigned Device Nickname
  - *Source Path:* Lab_4_1_1/deviceNameSharedPref.xml
  - *Raw Value:* <string name="deviceNameKey">Amazing Assistant</string>
  - *Qualification & Limitation:* Identifies user-assigned device label; does not prove active user identity.
- **Finding F-03:** Wireless Access Point Configuration
  - *Source Path:* Lab_4_1_3/WifiConfigStore.xml
  - *Raw Value:* SSID: Fios-pJK9q, Key: cop576him662ill
  - *Qualification & Limitation:* Establishes local wireless network binding and WPA keys; does not prove physical user location.
- **Finding F-04:** Bluetooth Peripheral Pairing
  - *Source Path:* Lab_4_1_3/bt_config.conf
  - *Raw Value:* Target: Sarfraz's MacBook Pro (MAC: f4:5c:89:c7:54:f0)
  - *Qualification & Limitation:* Identifies paired peripheral device pairing keys; requires physical proximity context.
- **Finding F-05:** User Security Restrictions
  - *Source Path:* Lab_4_1_2/userlist.xml
  - *Raw Value:* no_sms="true", no_outgoing_calls="true"
  - *Qualification & Limitation:* Confirms guest security policies enforced on primary system User ID 0.
- **Finding F-06:** Web Browser History Verification
  - *Source Path:* Lab_4_2_1/History (256 KB)
  - *Raw Value:* SQLite format 3
  - *Qualification & Limitation:* Manual binary inspection confirmed valid SQLite database header and visits table structure.
- **Finding F-07:** SIP Voice Communication Session
  - *Source Path:* Lab_4_2_2/chatgptJSON.json
  - *Raw Value:* Call ID 5c32191e-..., Callee: "Hibna Sarfraz"
  - *Qualification & Limitation:* Documents VoIP call parameters and participant URI mapping via Bacon framework.
- **Finding F-08:** Extraction Audit Limitation
  - *Source Path:* __All_Errors.txt

- *Raw Value:* Missing 24 carved JPEGs & XML nodes (xml-0-3, xml-0-8)
- *Qualification & Limitation:* Formally records dataset extraction limitations in compliance with ISO/IEC 27037.

8. Academic Integrity Declaration

I confirm that this report reflects my own examination of the authorized course evidence and that I have identified all sources, tools, and assistance used.

Student Name: Mohammad Muzamil

Registration Number: C11/26/DFIT/17289

Cohort: DFIT-COHORT-11

Submission Date: October 18, 2026